

$$R_n = \frac{\min(\delta_n, \delta_{n+1})}{\max(\delta_n, \delta_{n+1})}, \quad \delta_n = \varepsilon_{n+1} - \varepsilon_n$$

$$\hat{H}_{\text{dis}} = \hat{H}_{\text{flat}} - \sum_{a=1}^2 \sum_{y=0}^{N_y-1} \sum_{\mu=0}^1 v_{a,y} \hat{c}_{x_a,y,\mu}^\dagger \hat{c}_{x_a,y,\mu}, \quad (x_1, x_2) = (5, 15)$$

$$v_{a,y} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, W^2), \quad \mathbb{E}[v_{a,y}] = 0, \quad \mathbb{E}[v_{a,y} v_{b,y'}] = W^2 \delta_{ab} \delta_{yy'}$$

Disorder only at domain walls; independent along  $y$  and between walls; same for both orbitals.

$W$ : standard deviation;  $W^2$ : variance. Panels:  $W = 0.1, 0.2, 0.4$ .

$20 \times 32$ ; 100 realizations each; half system;  $y_0 = 0$ ;  $L = 0.99$

